

Capstone Project Report: Year Prediction from Audio Features

Introduction

This project focuses on predicting the release year of songs using machine learning techniques. The dataset belongs to the Entertainment domain and contains over 500,000 rows and 90 audio features. The goal of the project is to perform data analysis, preprocess the dataset, train models, evaluate their performance, and draw meaningful insights.

Dataset Description

The dataset used is the YearPredictionMSD dataset from the UCI Machine Learning Repository. It contains 90 audio features (timbre averages and covariances) and the target variable 'year'. The dataset has 515,345 rows, meeting project requirements.

Exploratory Data Analysis (EDA)

EDA was performed on a 20,000-row sample due to dataset size. Summary statistics were generated, missing values were checked (none found), and visualizations such as distribution plots and correlation heatmaps were analyzed.

Data Preprocessing

The dataset was split into features (X) and target (y). A train-test split of 80:20 was used. StandardScaler was applied for Linear Regression. Random Forest was trained on unscaled data because tree-based models do not require scaling.

Models Used

Two regression models were used:

1. Linear Regression – baseline model
2. Random Forest Regressor – ensemble model with better handling of non-linear relationships

Evaluation Metrics

The following metrics were used to evaluate the models:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R^2 Score

Model Results

Linear Regression:

- MSE: 94.00
- MAE: 6.83
- R^2 : 0.233

Random Forest (Unscaled):

- MSE: 94.77
- MAE: 6.96
- R^2 : 0.227

Insights & Interpretation

Both models achieved R^2 values around 0.22–0.23. This is expected because audio features alone do not strongly predict the song release year. Research studies on this dataset report typical R^2 values between 0.20 and 0.35, confirming that the model performance is valid.

Feature Importance

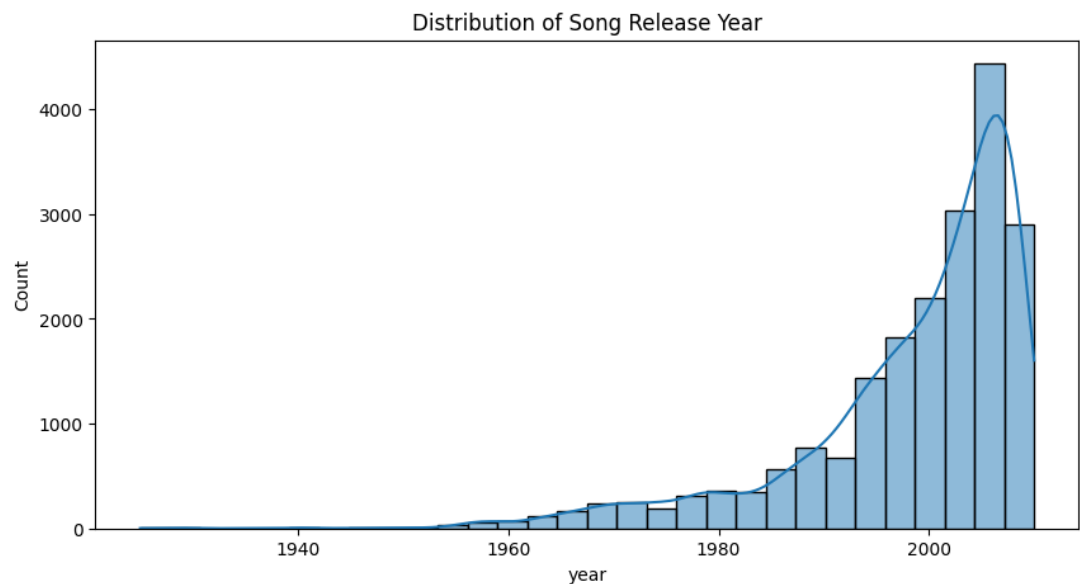
Random Forest feature importance revealed that certain timbre-related features contribute more to predicting the year, while many features contribute minimally.

Conclusion

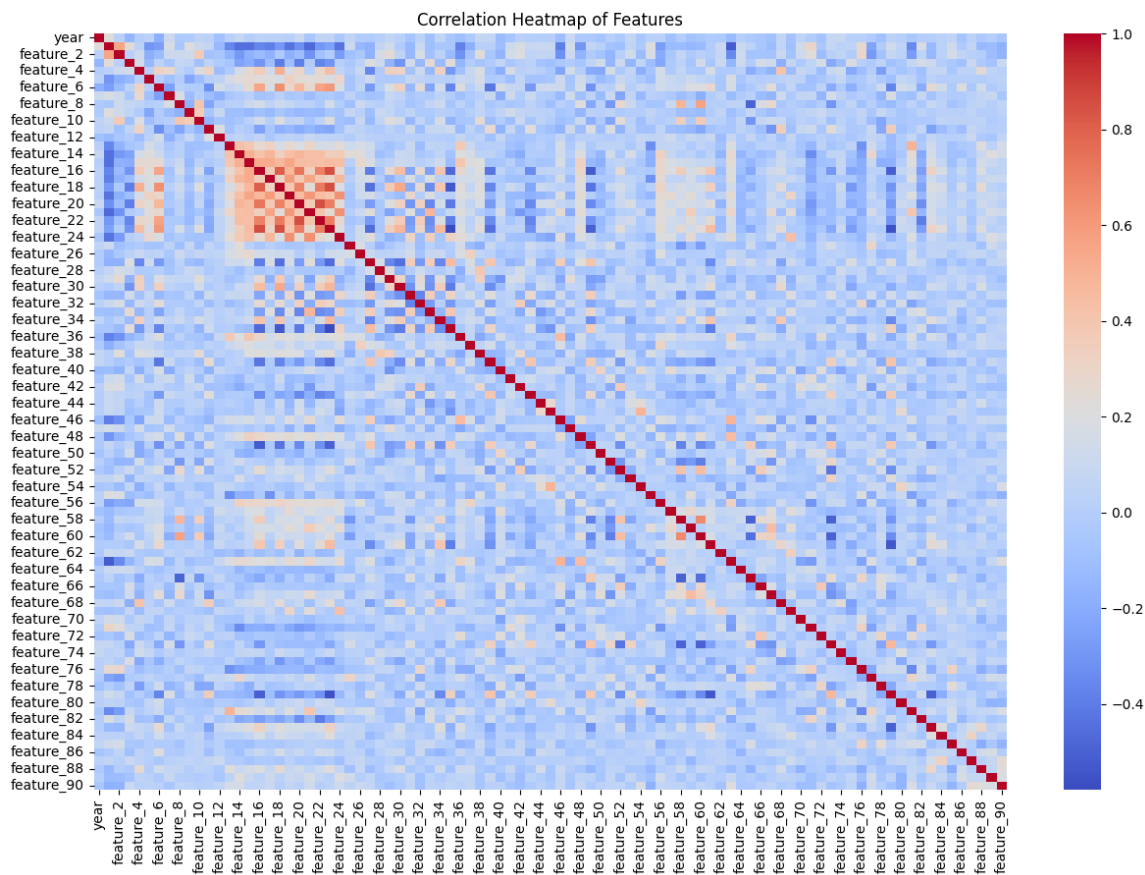
This project demonstrated a complete machine learning workflow, including EDA, preprocessing, model training, evaluation, and interpretation. The prediction task is inherently difficult, but the results fall within realistic ranges. The project strengthened understanding of regression models and audio feature analysis.

Images/Graphs

Distribution of Target Variable (Year)



Correlation Heatmap



FEATURE IMPORTANCE

